

# Internship experience report

Internship at Eviden (Atos Group)

Location: Madrid (Spain)

Master in Quantum Information Science and Technology

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This report has as purpose to describe my experience doing a 10-week internship at Eviden, as part of my master's programme in Quantum Information Science and Technology. Eviden is a subsidiary of Atos focused on sustainable digital solutions in areas such as Cybersecurity, High performance computing, Artificial Intelligence and Quantum Computing. The main objective of the internship was to develop a Quantum Machine Learning (QML) library within Eviden's software development stack, myQLM.

In the following, I will describe Eviden's Quantum business solutions. After this introduction, the problem statement of the project will be outlined, along its impact on the organization. Finally, I will reflect on my overall experience working in the company.

## 1 Eviden Quantum

Eviden offers different products and services within Quantum Computing. First of all, as mentioned early, it provides free access to myQLM, a Python package that allows students and developers to test their ideas directly from their laptops. In addition, it contains interoperability connectors with frameworks such as Qiskit, Cirq, ProjectQ or Forest, which gives flexibility to the user [1]. Finally, it can be used as a front-end to the Qaptiva 800, a quantum emulator that offers higher performances, larger simulation capabilities, being able to simulate up to 40 logical qubits, and advanced features like quantum circuit optimizers and noisy simulators [2].

Additionally, Eviden offers Qaptiva HPC, a licensed software that offers a full programming and emulation environment specifically designed for high-performance computing (HPC) environments. This innovative solution significantly boosts simulation capabilities, resulting in remarkable improvements in computing speed, with capacity limited only by the cluster memory. It is built in C++ with a Python API, enabling seamless integration and usability. Finally, it is designed for multi-node operation, being fully compatible with Slurm scheduling [3].

Some of the clients of Eviden in the past years have been the Jülich Supercomputing centre (Germany) [4], CESGA (Spain) [5], SCAYLE (Spain) [6], Hartree centre (UK) [7], GENCI (France) [6] and Université de Reims Champagne-Ardenne (France) [6]; and it has participated in big European projects such as HPCQS [8] and NEASQC [9].

Apart from selling the products mentioned before, assessing use cases and providing support, Eviden also delivers training workshops in universities. The workshops serve as an introduction to the fundamentals concepts of quantum computing, as well as some more advanced quantum algorithms. Furthermore, the workshops include applying the concepts learned with the help of Jupyter Notebook tutorials [10].

## 2 Problem statement

Quantum Machine Learning is a very popular area of research nowadays, in both academia and industry. Some of the giants of quantum computing such as IBM, Xanadu or Google have already developed a QML library, as well as several tutorials of QML algorithms. However, Eviden did not have entered yet into this field, even though they do have many quantum optimization classes implemented, as well as many ongoing research projects. Thus, I was appointed to develop a QML library in myQLM, together with Jupyter Notebook tutorials applying the algorithms to specific datasets.

This would bring value to Eviden in several ways. First of all, it would allow users to explore QML methods from a high-level perspective. Secondly, it would make myQLM more competitive with respect to other quantum SDKs. Thirdly, it would enable the delivery of more advanced training modules in educational quantum computing workshops. Fourthly, it would motivate the community, including researchers, to create and test their own QML routines, and develop demos which could be added to the official myQLM repository [11]. Finally, my work would serve to assess the performance of state of the art quantum machine learning algorithms.

## 3 Reflection

Reflecting back on my internship, I strongly believe it was a very valuable experience. First of all, I delved into the topic of Quantum Machine Learning, which I knew close to nothing in the beginning, and even learnt new classical Machine Learning algorithms, optimizers, and methods such as PCA. Secondly, I further improved my Python object-oriented programming (OOP) skills, learning useful concepts for package development such as abstract classes, overriding and inheritance. Thirdly, thanks to the interaction with my team members, I discovered which business activities the company works on, and that there is a lot of work to gain projects, including being in contact with clients and thinking strategically. Finally, through the progress presentations to the team, I improved my ability to explain concepts and results clearly, as well as to decide which information is relevant to communicate depending on the audience.

This internship met my expectations, as I had already done an internship in this same department during the last year of my Bachelor's. Moving from the learning outcomes to the work atmosphere, I can ensure that Eviden Quantum is a small team, but everyone in the team was friendly and helpful. In particular, my supervisor Andrés was very committed, gave me freedom to explore different options as well as confidence. He never micromanaged me, but also was attentive to ask me if I was stuck, suggesting possible solutions of research paths. Therefore, when I graduate, Eviden is going to be in my list of options, and I will have the possibility to work in multiple countries, as Eviden has offices in most European countries.

## References

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